A Tutorial on Support Vector Machines Using the Iris Dataset: Kernel Comparison and Classification Performance

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https://github.com/gondyalaakshayasree/individual-assignment

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Abstract

In many different sectors, machine learning techniques are routinely applied to categorization challenges. This lesson focuses on using the Iris dataset to use Support Vector Machines (SVM) for multi-class classification. The goal is to illustrate how SVM models work, how various kernel functions affect classification performance, and how standard metrics can be used to evaluate models. SVM models with linear, polynomial, and radial basis function (RBF) kernels are trained and tested using the Iris dataset, which has four numerical characteristics and three classes. The findings demonstrate that while more sophisticated kernels do not considerably enhance performance, the linear SVM achieves perfect classification accuracy. Model behavior is shown using visualizations like ROC curves, confusion matrices, and decision boundaries. The results show that when the dataset is almost linearly separable, simpler models can perform better than more sophisticated ones. For students looking to successfully apply machine learning approaches, this lesson offers an organized and useful grasp of SVM.

1. Introduction

In contemporary applications, machine learning has emerged as a crucial tool for data analysis and prediction. Support Vector Machines (SVM) are a well-known machine learning technology that works well for classification tasks (Cortes and Vapnik, 1995). SVM separates data points from distinct classes by identifying the best decision boundary. In order to increase generalization on unseen data, the objective is to maximize the margin between classes.

The kernel function selection has a significant impact on SVM performance. SVM can handle both linear and non-linear correlations by using kernels to convert data into higher-dimensional spaces (Bishop, 2006). Radial basis function (RBF), polynomial, and linear are common kernel types that offer varying degrees of model complexity.

The objective of this lesson is to show how SVM may be used on an actual dataset, analyze the effects of various kernels, and assess model performance using suitable metrics. The Iris dataset was chosen because it is easy to use and appropriate for classification tasks. With the use of code, visuals, and analytical insights, the lesson aims to give a step-by-step explanation.

2. Dataset Description

Originally presented by Fisher in 1936, the Iris dataset is a well-known dataset in machine learning. The 150 samples are split equally into three classes: virginica, versicolor, and setosa. There are four numerical characteristics in every sample:

- Sepal length
- Sepal width
- Petal length
- Petal width

With 50 examples of each class, the dataset is balanced, preventing any class from controlling the model training procedure. Because they prevent bias in classification algorithms and enable more trustworthy evaluation metrics, balanced datasets are crucial (Han et al., 2011).

While versicolor and virginica show some overlap, exploratory research reveals that the setosa class is distinct from the other two classes. Because of this feature, the dataset can be used to assess both linear and non-linear classification methods.

3. Methodology

3.1 Data Preprocessing

The dataset is split 80:20 into training and testing sets prior to model training. To ensure that the distribution of classes in both sets is equal, stratified sampling is used. To guarantee that every feature contributes equally to the model, feature scaling is carried out utilizing standardization. Because SVM depends on distance-based calculations, this stage is especially crucial (Géron, 2019).

3.2 Support Vector Machine Model

A supervised learning algorithm for classification and regression applications is the Support Vector Machine. Finding a hyperplane that divides various classes with the largest margin is the main goal of SVM (Cortes and Vapnik, 1995).

Kernel functions are used to map non-linearly separable data into higher-dimensional space. Three kernels are assessed in this tutorial:

- Linear kernel – suitable for linearly separable data
- Polynomial kernel – introduces non-linear decision boundaries
- RBF kernel – captures complex non-linear relationships

Each model is trained using a pipeline that includes feature scaling and classification.

3.3 Model Evaluation Metrics

To evaluate model performance, several metrics are used:

- Accuracy – proportion of correctly classified instances
- Confusion Matrix – shows correct and incorrect predictions
- ROC Curve and AUC – measure classification performance across thresholds

These metrics provide a comprehensive understanding of how well the model performs (Sokolova and Lapalme, 2009).

4. Results and Analysis

4.1 Class Distribution

As shown in Figure 1, the dataset is perfectly balanced, ensuring unbiased model training. Every class is equally represented in the dataset, which is fully balanced. This guarantees that there is no bias in the evaluation outcomes toward any class.

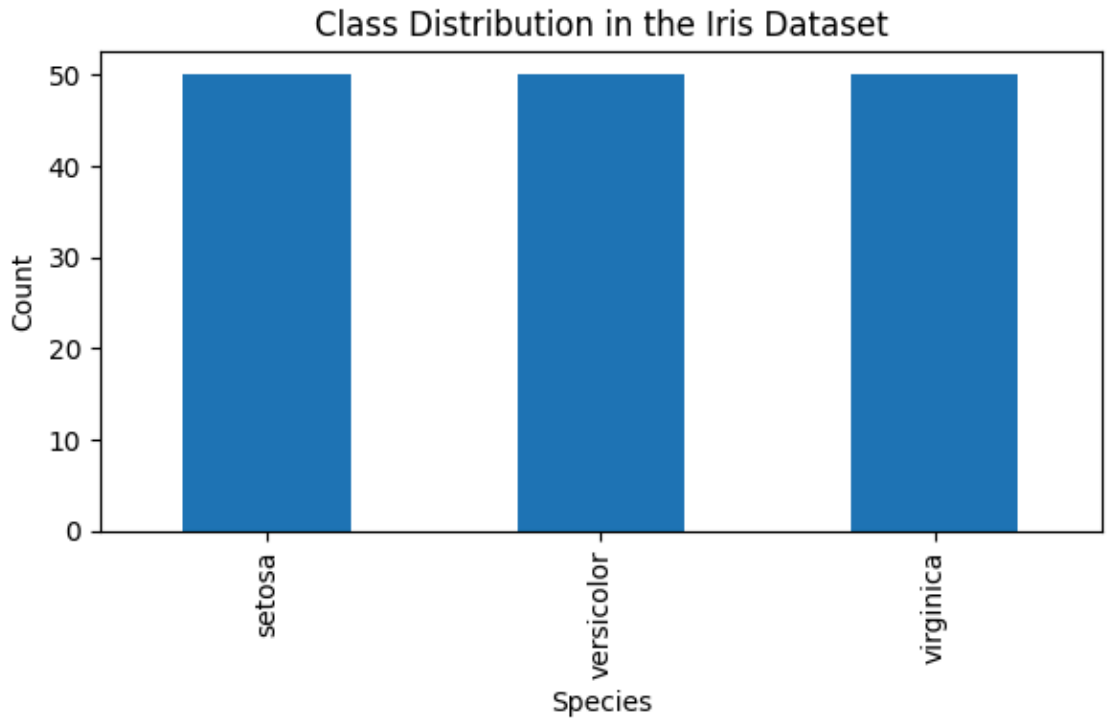

Figure 1: Class distribution of the Iris dataset showing equal representation of all three classes.

4.2 Feature Analysis

Figure 2 shows that setosa is clearly separable, while versicolor and virginica overlap. Setosa is distinct from the other classes, as seen by scatter plots. In feature space, however, versicolor and virginica overlap. This suggests that there is some linear separability in the dataset.

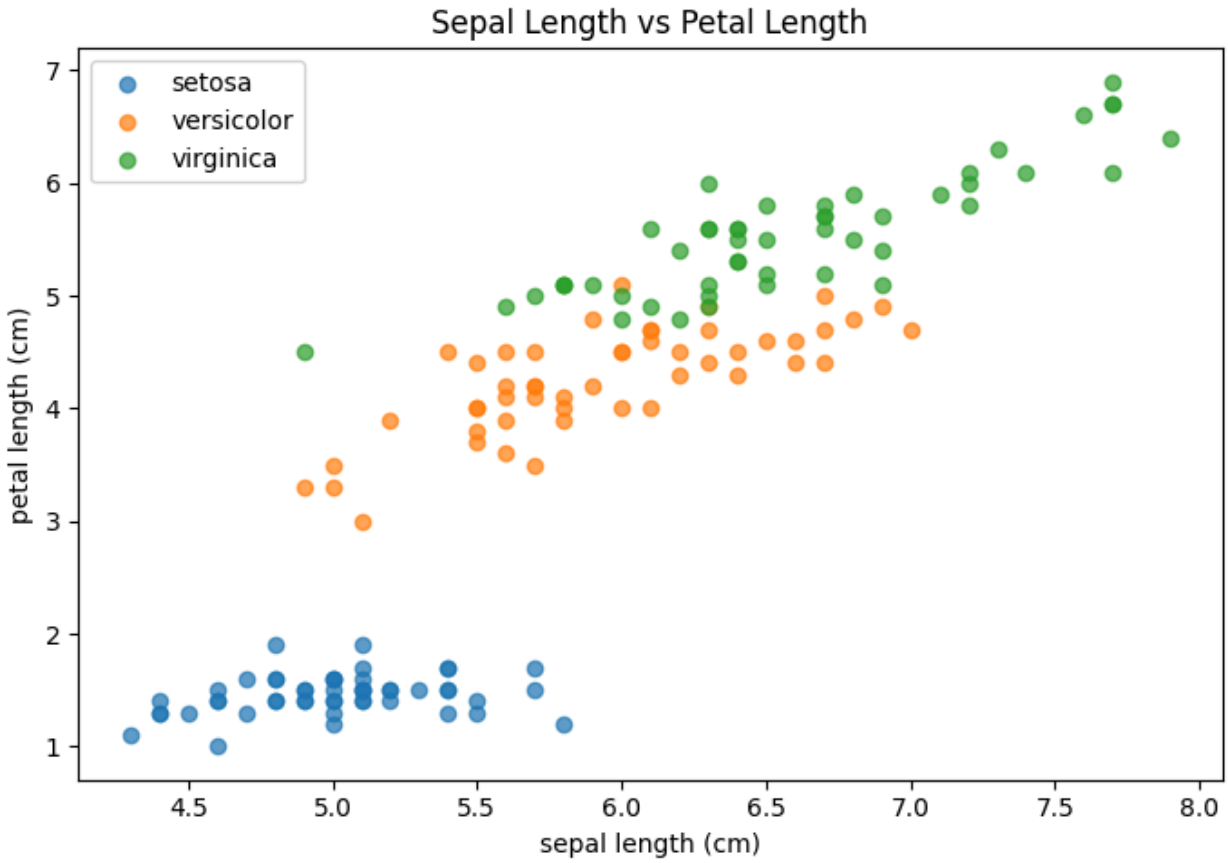

Figure 2: Scatter plot illustrating the relationship between sepal length and petal length across different species.

4.3 Kernel Performance Comparison

As illustrated in Figure 3, the linear kernel achieves the highest accuracy. The performance of different kernels is summarised below:

- Linear SVM: 100% accuracy
- Polynomial SVM: 90% accuracy
- RBF SVM: 97% accuracy

The dataset is mostly linearly separable since the linear SVM achieves perfect classification. The polynomial kernel performs the worst, most likely as a result of improper feature space transformation or overfitting. Although it works well, the RBF kernel falls short of the linear model.

This illustrates that performance does not always increase with model complexity.

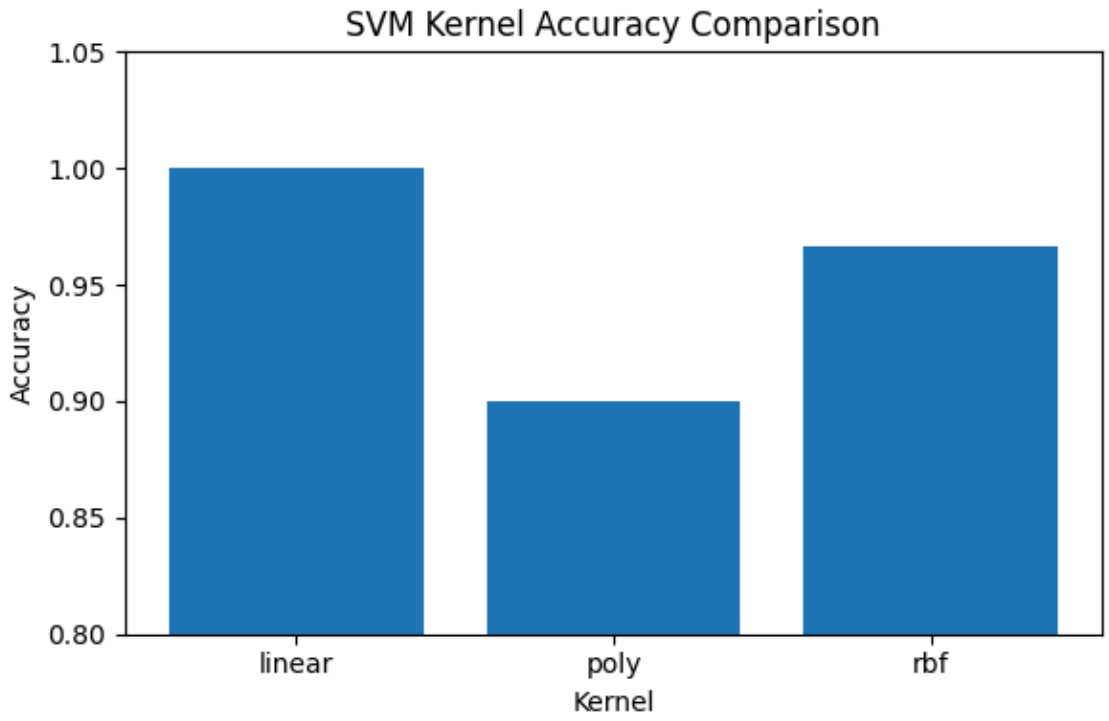

Figure 3: Comparison of classification accuracy across different SVM kernels.

4.4 Confusion Matrix Analysis

Figure 4 confirms that all predictions were correctly classified. With every occurrence accurately predicted, the linear SVM's confusion matrix demonstrates flawless classification. Strong model performance and outstanding generalization are indicated by this.

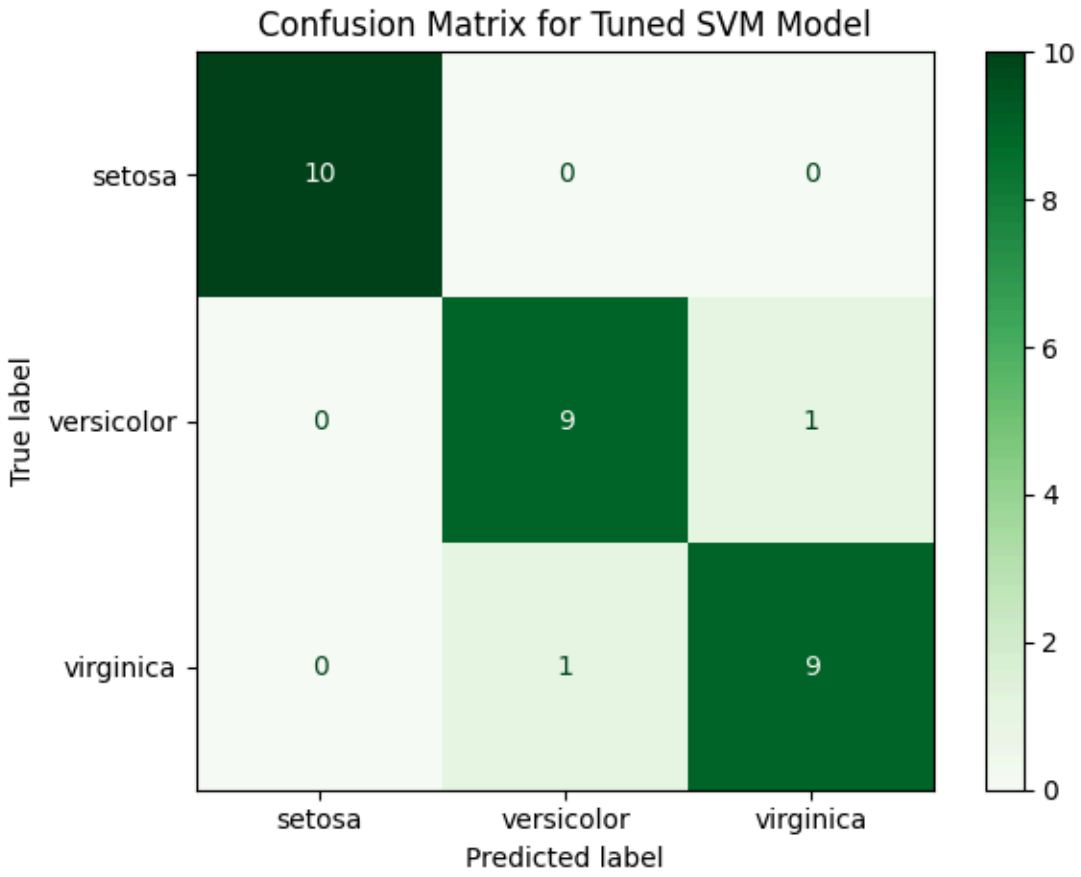

Figure 4: Confusion matrix for the tuned SVM model showing minor misclassifications.

There are slight misclassifications between versicolor and virginica for the tweaked SVM model. Because these classes overlap in the feature space, this is to be expected.

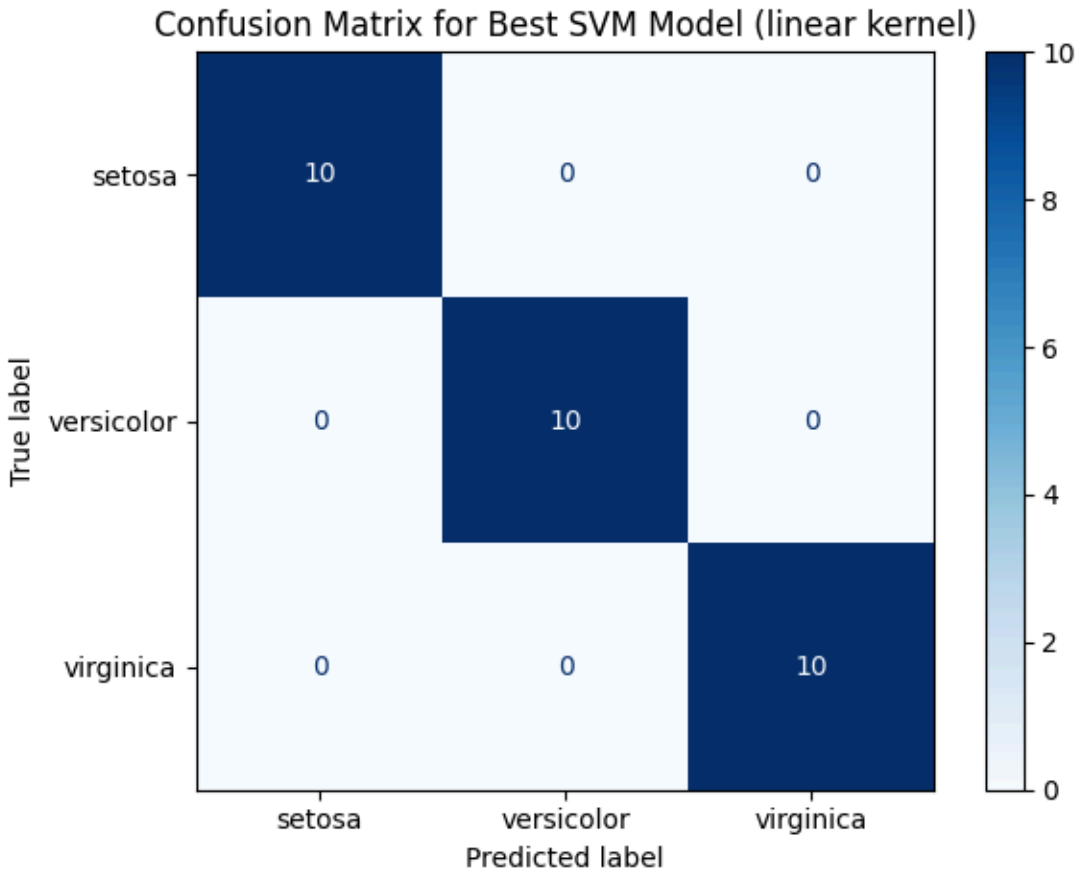

Figure 5: Confusion matrix showing perfect classification performance of the linear SVM model.

4.5 PCA Visualisation

Figure 6 shows clear separation of setosa and overlap between other classes. The dataset is reduced to two dimensions for visualization using Principal Component Analysis (PCA). Setosa forms a discrete cluster, whereas the other two groups are closer together, according to the PCA plot. This validates the partial overlap that was previously noted.

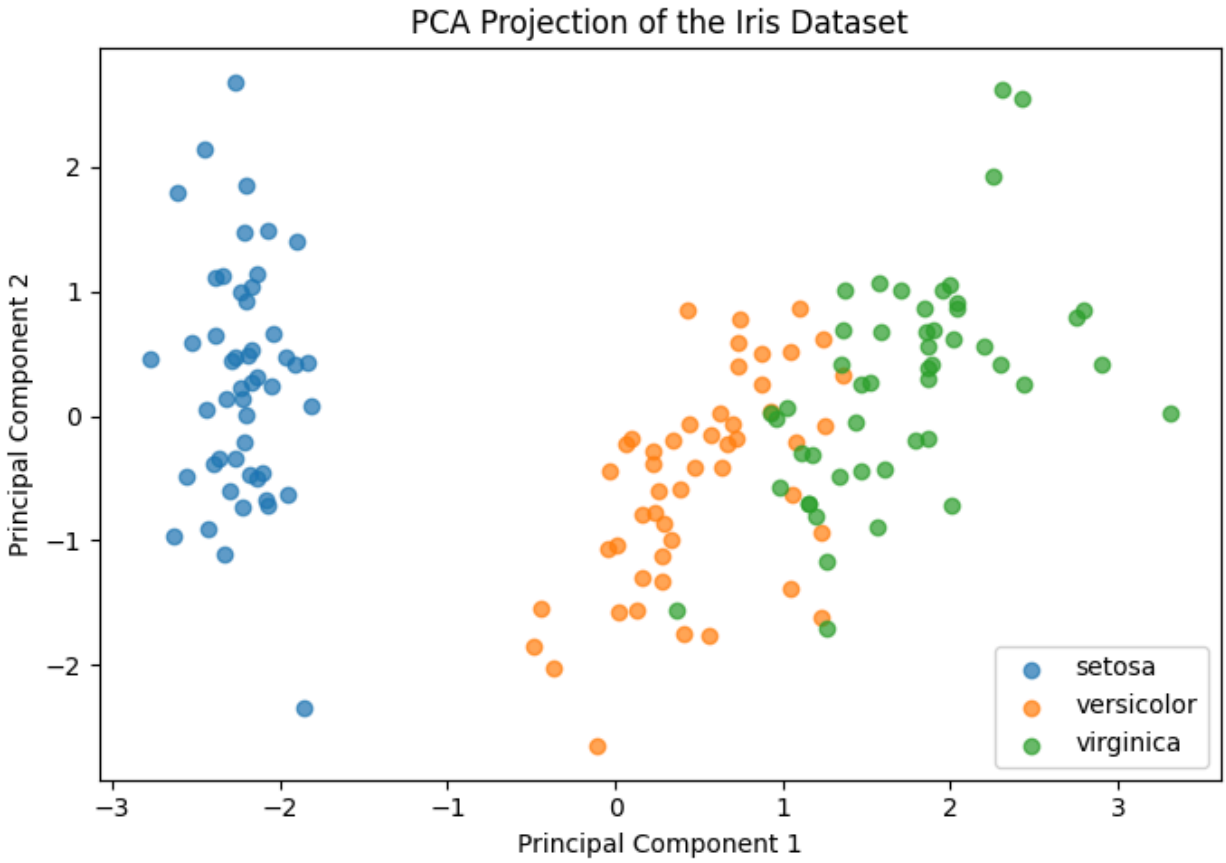

Figure 6: PCA projection showing clustering of Iris classes in reduced dimensions.

4.6 Decision Boundary Analysis

As illustrated in Figure 7, the SVM creates clear separation boundaries between classes. The decision boundary map demonstrates how well the SVM divides the classes. The distinction between versicolor and virginica is less obvious than that between setosa and virginica.

This demonstrates how SVM can handle overlapping data while maximizing the margin between classes.

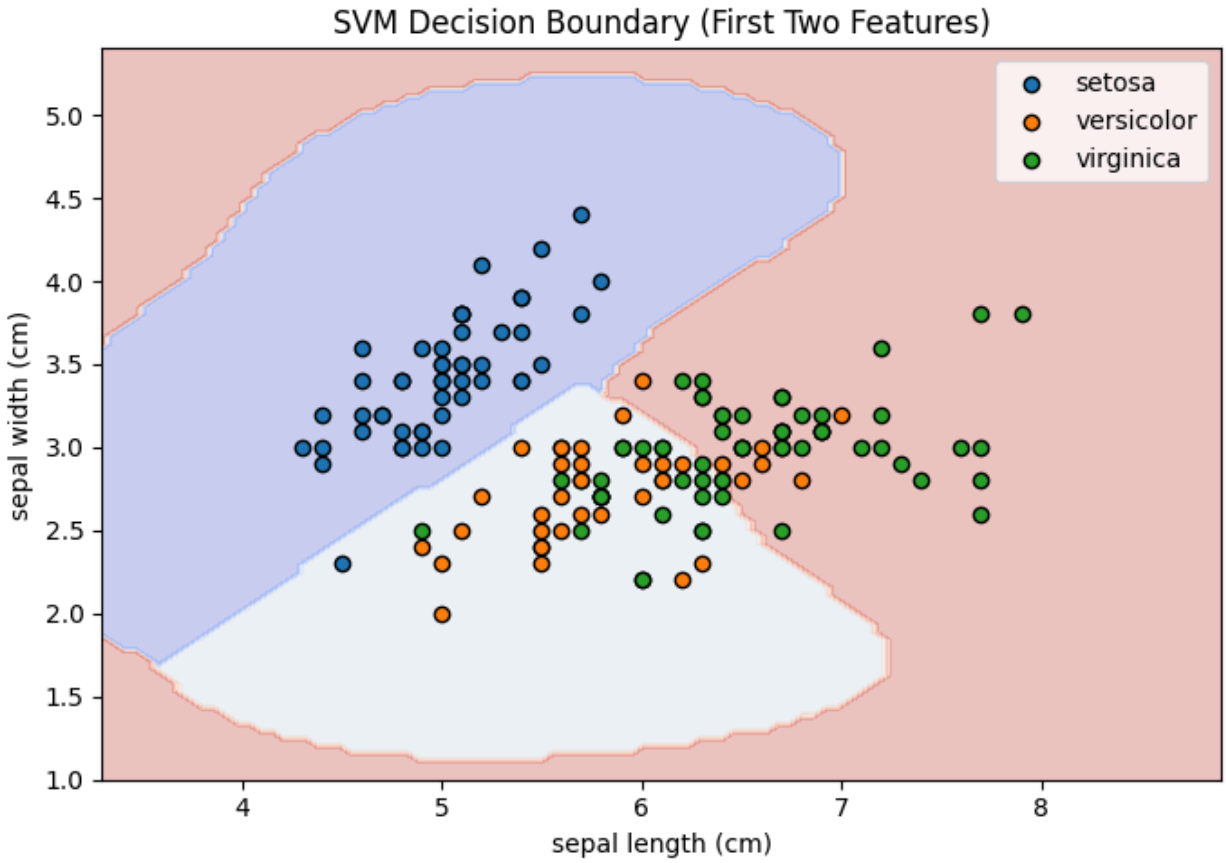

Figure 7: Decision boundary generated by the SVM model using the first two features.

4.7 ROC Curve Analysis

Figure 8 demonstrates high AUC values, indicating strong model performance.

The ROC curves show excellent classification performance:

- Setosa: AUC = 1.00
- Versicolor: AUC ≈ 0.98
- Virginica: AUC ≈ 0.98

High AUC values indicate strong discriminative ability. The perfect AUC for setosa confirms its clear separability, while slightly lower values for the other classes reflect their overlap.

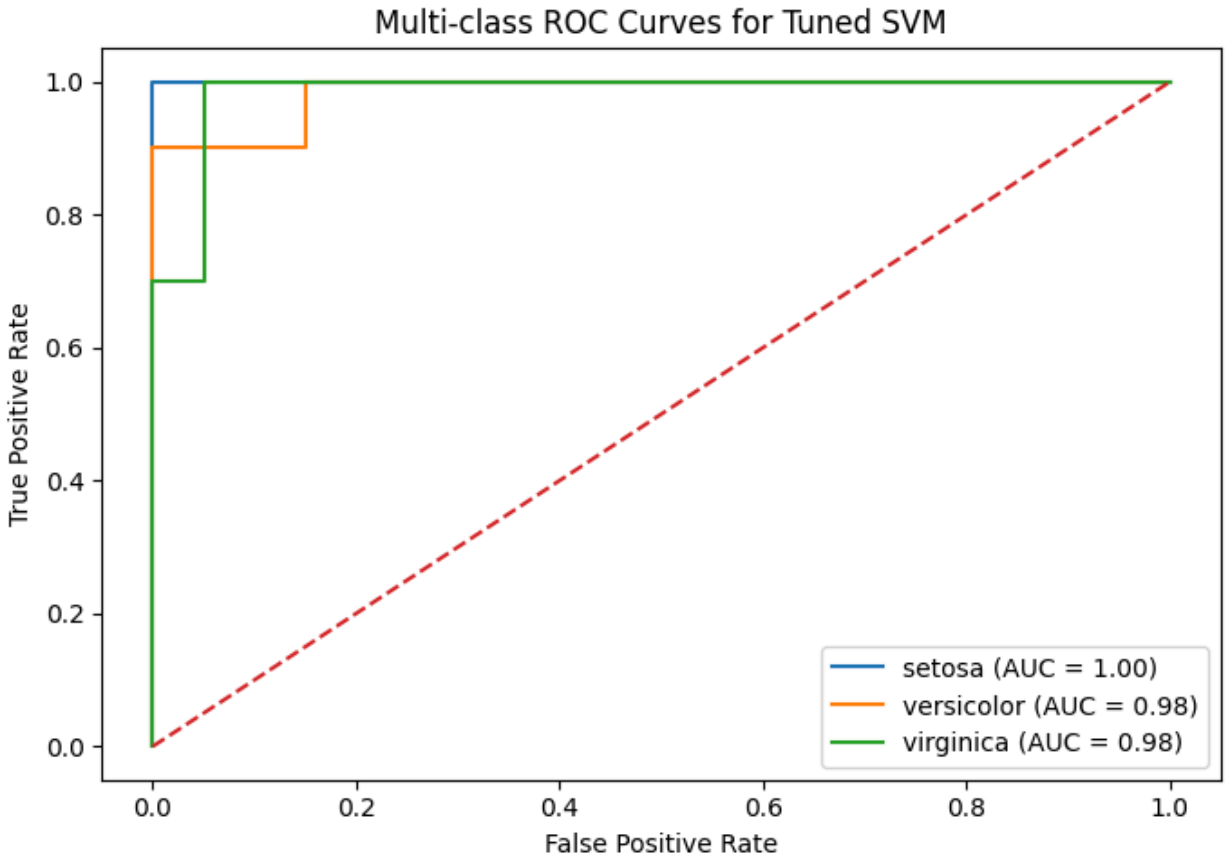

Figure 8: ROC curves for each class showing high classification performance.

4.8 Final Model Comparison

As shown in Figure 9, the linear SVM outperforms other models. The linear SVM performs the best when all models are compared. It's interesting to note that hyperparameter adjustment marginally decreases accuracy and has no effect on performance.

This implies that simpler methods work better, and that sophisticated modeling of the dataset is not necessary.

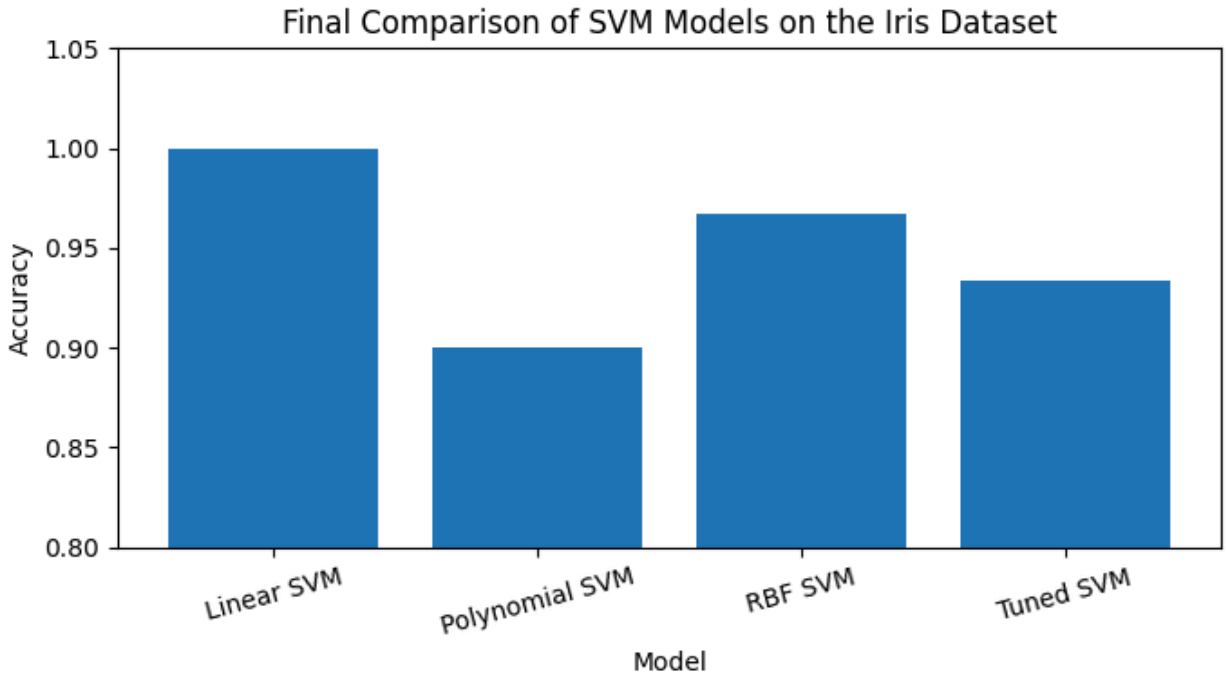

Figure 9: Final comparison of SVM models highlighting the best-performing approach.

5. Critical Evaluation

The findings show that SVM is an effective classification algorithm, especially for datasets with distinct class boundaries. When the data is linearly separable, the linear SVM achieves 100% accuracy, demonstrating its efficacy.

Nevertheless, the performance of more sophisticated kernels like polynomial and RBF does not considerably enhance outcomes. The polynomial kernel actually performs worse, suggesting possible overfitting. This highlights how crucial it is to choose a suitable model based on the features of the dataset.

SVM's sensitivity to feature scale and parameter selection is another drawback. Inadequate tuning could result in less-than-ideal performance. Furthermore, SVM may become computationally costly for larger datasets, even when it works well on smaller ones like Iris (Bishop, 2006).

SVM is still a strong and dependable algorithm in spite of these drawbacks. Effective model interpretation is supported, and comprehension is improved by the use of visualization techniques like PCA and decision boundaries.

6. Conclusion

Using the Iris dataset, this tutorial illustrates how Support Vector Machines can be used for multi-class classification. The findings demonstrate that while more complicated kernels do not significantly enhance accuracy, the linear SVM achieves perfect accuracy.

The research shows that the model's complexity should match the features of the dataset. When the data is well-structured, simpler models frequently perform better than more complex ones. Evaluation methods like ROC curves and confusion matrices verify the model's efficacy.

All things considered, SVM turns out to be a strong and adaptable method for classification applications. This tutorial is appropriate for beginning and advanced learners because it offers a clear and useful explanation of how SVM operates.

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